Interpolating zeros force nonseparability in the de Branges gap

Autonomous solution packet for arXiv:0906.2943

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Status: candidate partial result, likely valid. If the zeros of $\Theta = E^\#/E$ contain a Riesz kernel subsequence of divergent total imaginary part, then the imaginary-half-line majorant space contains a bicontinuous copy of ℓ^∞. Thus it is neither separable nor reflexive. Highly clustered or multiple-zero configurations remain open.

1 Source question and the unresolved gap

Let $\mathcal{H} = \mathcal{H}(E)$ be a de Branges space, put $\Theta = E^\#/E$, and define on $D = i[1,\infty)$

$$\mathfrak{m}_E(z) = \frac{|E(z)|}{|z+i|}.$$

The Banach space $R_\mathfrak{m}(\mathcal{H})$ consists of all $F \in \mathcal{H}$ for which both $|F|$ and $|F^\#|$ are bounded by a constant multiple of $\mathfrak{m}_E$ on D, with norm

$$\|F\|_\mathfrak{m} = \max\left\{\|F\|_\mathcal{H}, \sup_{y\geq 1}\frac{|F(iy)|}{\mathfrak{m}_E(iy)}, \sup_{y\geq 1}\frac{|F^\#(iy)|}{\mathfrak{m}_E(iy)}\right\}. \tag{1}$$

Baranov and Woracek prove that $R_\mathfrak{m}(\mathcal{H}) = \mathcal{H}$ precisely when the mean type of Θ is zero and the sum of the imaginary parts of its zeros is finite. They also embed ℓ^∞ when the mean type is negative or the zero heights have positive limsup [1, Thms. 3.1–3.2]. Remark 3.3 asks whether the remaining gap can contain a proper $R_\mathfrak{m}(\mathcal{H})$ that is separable or reflexive.

The result below excludes every gap candidate whose zeros carry divergent height on a Riesz subsequence. This includes interpolating Blaschke zero sequences.

2 Proof intuition

At a zero v of Θ, the normalized de Branges kernel is, up to a unimodular constant, $E(z)\sqrt{\operatorname{Im} v}/(\bar{v}-z)$. A Cauchy coefficient λ therefore costs $|\lambda|^2/\operatorname{Im} v$ in squared Hilbert norm. Equal coefficients are too expensive when the zero heights tend to zero.

Instead, group a divergent-height Riesz subsequence into remote finite blocks. To encode an increment Δ_k, assign to a zero v in block k the coefficient

$$\lambda_v = \Delta_k \frac{\operatorname{Im} v}{H_k}, \qquad H_k = \sum_{v \text{ in block } k} \operatorname{Im} v.$$

The coefficients still sum to Δ_k, but their squared Hilbert cost is exactly $|\Delta_k|^2/H_k$. Taking $H_k \geq k^4$ makes the total cost finite. Large gaps between blocks make both relevant Cauchy transforms uniformly $O(1/y)$; between blocks, their completed sums recover the encoded bounded sequence. The formal proof makes these estimates quantitative.

3 A block Cauchy estimate

Lemma 1. *Let V_k be finite disjoint sets of points $v = x + ih$ in $\mathbb{C}^+$ with $h \le 1/2$ and $|v| \ge 2$. Write*

$$P_k = \min_{v \in V_k} |v|, \quad Q_k = \max_{v \in V_k} |v|, \quad A_k = \sum_{r \le k} Q_r.$$

Suppose the blocks are radially ordered and their starts can be chosen so large that

$$P_{k+1} \ge 10^6 A_k, \tag{2}$$

with the same condition imposed recursively after every block is chosen. For $d = (d_k) \in \ell^\infty$, set $d_0 = 0$, $\Delta_k = d_k - d_{k-1}$, and suppose coefficients λ_v satisfy

$$\sum_{v \in V_k} \lambda_v = \Delta_k, qquad \sum_{v \in V_k} |\lambda_v| \le 2\|d\|_\infty. \tag{3}$$

Then

$$\sup_{y \ge 1} y \left| \sum_{k, v \in V_k} \frac{\lambda_v}{\bar{v} - iy} \right| + \sup_{y \ge 1} y \left| \sum_{k, v \in V_k} \frac{\bar{\lambda}_v}{v - iy} \right| \le C\|d\|_\infty \tag{4}$$

for an absolute constant C. Moreover, with $y_k = (A_k P_{k+1})^{1/2}$,

$$\left| -iy_k \sum_{r, v \in V_r} \frac{\lambda_v}{\bar{v} - iy_k} - d_k \right| \le \frac{1}{4}\|d\|_\infty. \tag{5}$$

Proof. The elementary denominator bounds are

$$|\bar{v} - iy| \ge \max\{|v|, y\}, \qquad |v - iy| \ge c \max\{|v|, y\}, \tag{6}$$

where the second follows from $h \le 1/2$, $y \ge 1$, and $|v| \ge 2$. If $P_k \le y < P_{k+1}$, split the sums into completed blocks $r < k$, the current block, and future blocks. For completed blocks use

$$\frac{1}{\bar{v} - iy} = -\frac{1}{iy} + \frac{\bar{v}}{iy(\bar{v} - iy)}, \tag{7}$$

and the analogous identity with v in place of $\bar{v}$. The main term has modulus at most $\|d\|_\infty / y$. After multiplication by y, the error is bounded by $2\|d\|_\infty A_{k-1}/y$, hence uniformly by (2). The entire current block costs at most a constant times $2\|d\|_\infty / y$ by (3) and (6). The future blocks cost at most

$$2\|d\|_\infty \sum_{r > k} P_r^{-1} \lesssim \frac{\|d\|_\infty}{P_{k+1}},$$

because (2) makes the P_r grow geometrically. This proves (4) for $y \ge P_1$. If $1 \le y < P_1$, every block is a future block, so the same geometric estimate proves (4) directly.

At y_k, all blocks through k are completed. Equation (7) gives the main term d_k. The past error is at most $2\|d\|_\infty A_k / y_k$, and the future error after multiplication by y_k is at most a constant times $2\|d\|_\infty y_k / P_{k+1}$. Both are bounded by a fixed constant times

$$\|d\|_\infty \sqrt{A_k / P_{k+1}}.$$

The numerical factor 10^6 in (2), enlarged during the recursive choice if needed, makes their sum at most $\|d\|_\infty / 4$. This proves (5). □

4 Main partial theorem

Theorem 2. *Let $\mathcal{H}(E)$ be a de Branges space and let (v_n) be a subsequence of the zeros of $\Theta = E^{\#}/E$ in $\mathbb{C}^+$. Assume*

1. *the normalized kernels $\widetilde{K}(v_n,\cdot) = K(v_n,\cdot)/\|K(v_n,\cdot)\|_{\mathcal{H}}$ form a Riesz sequence in $\mathcal{H}(E)$;*
2. *$\sum_n \operatorname{Im} v_n = \infty$.*

Then $\langle R_{\mathfrak{m}}(\mathcal{H}), \|\cdot\|_{\mathfrak{m}}\rangle$ contains a bicontinuous copy of ℓ^∞. Consequently it is neither separable nor reflexive.

Proof. Discarding finitely many points, assume $h_n := \operatorname{Im} v_n \leq 1/2$ and $|v_n| \geq 2$. Since every tail still has divergent height, choose radially ordered finite blocks V_k satisfying the separation in Lemma 1 and

$$H_k := \sum_{v_n \in V_k} h_n \geq k^4. \tag{8}$$

This is done recursively: skip beyond the required next radius, then take a finite initial segment of the remaining divergent-height tail.

For $d \in \ell^\infty$, define, for $v_n \in V_k$,

$$\lambda_n = (d_k - d_{k-1})\frac{h_n}{H_k}. \tag{9}$$

Then (3) holds, and the decisive energy identity is

$$\sum_{v_n \in V_k} \frac{|\lambda_n|^2}{h_n} = \frac{|d_k - d_{k-1}|^2}{H_k}. \tag{10}$$

At a zero v_n of Θ, the normalized-kernel formula used in the proof of Proposition 3.8 of [1] reduces to

$$\frac{\widetilde{K}(v_n,z)}{E(z)} = \frac{\sqrt{h_n}}{i\sqrt{\pi}}\,\frac{\overline{E(v_n)}}{|E(v_n)|}\,\frac{1}{\bar{v}_n - z}. \tag{11}$$

Choose the coefficient of $\widetilde{K}(v_n,\cdot)$ so that its quotient by E in (11) is $\lambda_n/(\bar{v}_n - z)$, and let F_d be the resulting series. The coefficient modulus is $\sqrt{\pi}|\lambda_n|/\sqrt{h_n}$. The Riesz synthesis bound, (8), and (10) give

$$\|F_d\|_{\mathcal{H}}^2 \lesssim \sum_k \frac{|d_k - d_{k-1}|^2}{H_k} \leq 4\|d\|_\infty^2 \sum_k k^{-4}. \tag{12}$$

Thus the series converges in $\mathcal{H}$ and

$$\frac{F_d(z)}{E(z)} = \sum_{k, v_n \in V_k} \frac{\lambda_n}{\bar{v}_n - z}. \tag{13}$$

Taking sharp conjugates gives

$$\frac{F_d^{\#}(z)}{E(z)} = \Theta(z) \sum_{k, v_n \in V_k} \frac{\bar{\lambda}_n}{v_n - z}. \tag{14}$$

Since $|\Theta(iy)| \leq 1$, Lemma 1, (12)–(14), and compact point-evaluation bounds on $1 \leq y \leq P_1$ imply

$$\|F_d\|_{\mathfrak{m}} \lesssim \|d\|_\infty. \tag{15}$$

Indeed, division by $\mathfrak{m}_E(iy) = |E(iy)|/(y+1)$ turns the two majorization terms into $(y+1)$ times the two Cauchy transforms.

Finally, (5) and (13) yield

$$\|F_d\|_{\mathfrak{m}} \geq y_k \left|\frac{F_d(iy_k)}{E(iy_k)}\right| \geq |d_k| - \frac{1}{4}\|d\|_\infty.$$

Taking the supremum over k gives

$$\|F_d\|_{\mathfrak{m}} \geq \frac{3}{4}\|d\|_\infty. \tag{16}$$

Hence $d \mapsto F_d$ is linear, bounded, and bounded below from ℓ^∞ into $R_{\mathfrak{m}}(\mathcal{H})$. Its range is closed and bicontinuously isomorphic to ℓ^∞, proving the theorem. □

Corollary 3. *If the zeros of Θ contain an interpolating Blaschke subsequence of divergent total imaginary part, then $R_{\mathfrak{m}}(\mathcal{H})$ is neither separable nor reflexive. In particular, the zero sequence*

$$v_n = 2^n + \frac{i}{n} \qquad (n \geq 1) \tag{17}$$

defines a meromorphic-inner/de Branges example in the source's unresolved height gap for which $R_{\mathfrak{m}}(\mathcal{H})$ nevertheless contains ℓ^∞.

Proof. For an interpolating Blaschke sequence, normalized kernels at its zeros form a Riesz sequence. In (17), the Blaschke sum $\sum_n (1/n)/(1+4^n)$ converges, while $\sum_n 1/n$ diverges and the heights tend to zero. Uniform pseudohyperbolic separation follows from the exponential spacing of the real parts: for $m \neq n$,

$$1 - \left|\frac{v_n - v_m}{v_n - \bar{v}_m}\right|^2 = \frac{4\,\operatorname{Im} v_n \operatorname{Im} v_m}{|v_n - \bar{v}_m|^2},$$

and the sum of these defects tends to zero as $n \to \infty$; the finitely many initial products are positive. Thus the sequence is interpolating. Apply Theorem 2. The standard correspondence between meromorphic inner functions and Hermite–Biehler functions supplies E with $E^\#/E = \Theta$. □

5 Scope, verification, and novelty

This theorem closes a substantial part of Remark 3.3's gap but not all of it. A divergent zero-height sum can be concentrated in increasingly tight clusters or high multiplicities. Such a configuration need not visibly contain a Riesz subsequence retaining divergent height. Extending the block argument would require uniform frame estimates for derivative-kernel blocks; none is proved here.

The script `code/verify_weighted_blocks.py` checks the exact energy identity (10), both finite Cauchy-transform bounds, and recovery between five lacunary blocks over 80 deterministic random trials. It is a sanity check, not part of the proof.

A bounded novelty search on August 11, 2026 used the exact source question, the paper title, "interpolating zeros," "Riesz kernels," and "nonseparable majorant space." It found the Baranov–Woracek series and adjacent de Branges interpolation literature, but no matching theorem or full resolution. Novelty confidence is moderate, not exhaustive.

References

[1] A. Baranov and H. Woracek, *Majorization in de Branges spaces II. Banach spaces generated by majorants*, Collect. Math. 62 (2011), 27–55; arXiv:0906.2943.

Source evidence

Page 7 of arXiv:0906.2943 states the equality criterion, the known ℓ^∞-embedding regime, and the separability/reflexivity question in Remark 3.3.

$$\liminf_{\substack{|z|\to\infty \\ z\in\Gamma_\alpha}} \Big(|z|\cdot|e^{i\varphi}-\Theta(z)|\Big)<\infty\,.$$

(iii) We have $\operatorname{mt}\Theta=0$ *and* $\sum_{n\in\mathbb{N}}\operatorname{Im}w_n<\infty$.

(iii′) The domain of the multiplication operator $S_{\mathcal{H}}$ *in* $\mathcal{H}$ *is not dense.*

3.2 Theorem. *Let* $\mathcal{H}=\mathcal{H}(E)$ *be a de Branges space, and set* $\mathfrak{m}:=\mathfrak{m}_E|_{i[1,\infty)}$. *Moreover, denote* $\Theta:=E^{-1}E^\#$, *and let* $(w_n)_{n\in\mathbb{N}}$ *be the sequence of zeros of* Θ *in* $\mathbb{C}^+$ *listed according to their multiplicities. Then*

$$(i)\implies(ii)\implies(iii),$$

where (i), (ii), and (iii), are the following conditions:

(i) We have $\operatorname{mt}\Theta<0$ *or* $\limsup_{n\to\infty}\operatorname{Im}w_n>0$.

(ii) There exists $\delta>0$, *such that*

$$\liminf_{\substack{|z|\to\infty \\ \operatorname{Im}z\geq\delta}} |\Theta(z)|=0\,. \tag{3.2}$$

(iii) There exists a bicontinuous embedding of $\langle\ell^\infty,\|.\|_\infty\rangle$ *into* $\langle R_{\mathfrak{m}}(\mathcal{H}),\|.\|_{\mathfrak{m}}\rangle$. *In particular,* $R_{\mathfrak{m}}(\mathcal{H})$ *is neither separable nor reflexive.*

3.3 Remark. Comparing the condition *(iii)* of Theorem 3.1 with *(i)* of Theorem 3.2, it is apparent that these theorems do not establish a full dichotomy; there is a gap between the described situations. At present it is not clear to us what happens in this gap. In particular, it is an open question whether there exists a de Branges space $\mathcal{H}=\mathcal{H}(E)$, such that for $\mathfrak{m}:=\mathfrak{m}_E|_{i[1,\infty)}$ the space $\langle R_{\mathfrak{m}}(\mathcal{H}),\|.\|_{\mathfrak{m}}\rangle$ is separable (or reflexive) although $R_{\mathfrak{m}}(\mathcal{H})\neq\mathcal{H}$.

7